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Chapter 2 Formula Sheet

1. Frequency & Period

$$f = \frac{1}{T}$$

$$T = \frac{1}{f}$$

- f = frequency in **Hz**
- T = period in **seconds**
- Frequency = cycles per second
- Period = time for one cycle

Your slides define frequency and period as inverses. Chapter 2

2. Bit Rate & Bit Interval

$$\text{Bit Interval} = \frac{1}{\text{Bit Rate}}$$

$$\text{Bit Rate} = \frac{1}{\text{Bit Interval}}$$

- Bit rate = **bits sent per second (bps)**
- Bit interval = **time needed to send one bit**

Example from your slides:

$$2000 \text{ bps} \rightarrow \frac{1}{2000} = 0.0005s = 500\mu s$$

3. Propagation Time

This is **how long one bit takes to travel through the link.**

$$\text{Propagation Time} = \frac{\text{Distance}}{\text{Propagation Speed}}$$

Usually:

- Distance $\rightarrow$ **meters (m)**
- Speed $\rightarrow$ **m/s**
- Answer $\rightarrow$ **seconds**

The slides note that propagation speed depends on the medium; light is about 3×10^8 m/s in vacuum and slower in cables.

4. Transmission Time

This is how long it takes to **push the entire message onto the link.**

$$\text{Transmission Time} = \frac{\text{Message Size in bits}}{\text{Bit Rate}}$$

So remember:

Propagation $\rightarrow$ distance/speed

Transmission $\rightarrow$ message size/bit rate

Your slides explicitly compare these two quantities in their examples.

5. Bandwidth–Delay Product

$$\text{Bandwidth} - \text{Delay Product} = \text{Bandwidth} \times \text{Propagation Delay}$$

Unit: *bits*

Meaning: **how many bits can fill the link at one time.**

Think of a pipe:

Bandwidth = how wide the pipe is

Delay = how long the pipe is

Bandwidth-delay product = how much the pipe can hold

6. Wavelength

$$\lambda = \text{Propagation Speed} \times T$$

Since:

$$T = \frac{1}{f}$$

you can also use:

$$\lambda = \frac{\text{Propagation Speed}}{f}$$

- λ = wavelength
- f = frequency
- Wavelength = distance occupied by **one cycle**.

Conversions You Need

These are especially important because this is where you've been making most of your small mistakes.

Prefix	Value
milli (m)	10^{-3}
micro (μ)	10^{-6}
nano (n)	10^{-9}
kilo (k)	10^3
mega (M)	10^6
giga (G)	10^9

And:

$$1 \text{ Byte} = 8 \text{ bits}$$

So, for example:

$$5 \text{ MB} = 5 \times 10^6 \text{ Bytes} \times 8 \text{ when a formula requires the message size in bits.}$$

Your conversion rule

The rule you've been using is good:

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Smaller units $\rightarrow$ bigger numerical value

Bigger units $\rightarrow$ smaller numerical value

For example:

$$2 \text{ MHz} = 2,000,000 \text{ Hz}$$

but:

$$2 \mu\text{s} = 0.000002 \text{ Hz}$$